

The effectiveness of intelligent tutoring systems on K-12 students' reading comprehension: A meta-analysis

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Abstract

This meta-analysis examined the effectiveness of improving reading comprehension for students in K-12 classrooms using intelligent tutoring systems (ITSs), a computer-based learning environment that provides customizable and immediate feedback to the learner. Nineteen studies from 13 publications incorporating approximately 10 000 students were included in the final analysis; using robust variance estimation to account for statistical dependencies, the 19 studies yielded 88 effect size estimates. The meta-analysis indicated that the overall random effect size of ITSs on reading comprehension was 0.60 (using a mix of standardized and researcher-designed measures) with a 95% confidence interval 0.36 to 0.85 ($p < 0.001$). This review confirms previous studies comparing ITSs to human tutoring: ITSs produced a small effect size when compared to human tutoring (0.20, 0.02–0.38, $p = 0.036$, $n = 21$). All comparisons to human tutoring used standardized measures. This review also found that ITSs produced a larger effect size on reading comprehension when compared to traditional instruction (0.86) for mixed measures and (0.26) for standardized measures. These findings may be of interest to practitioners and policy makers seeking to improve reading comprehension using consistent and accessible ITSs. Recommendations for researchers include conducting studies to understand the difference between traditional and updated versions of ITSs and employing valid and reliable standardized tests and researcher-designed measures.

Practitioner Notes

What is already known about this topic

- ITSs have a moderate positive effect on students' academic learning.
- ITSs produce a small effect size when compared to human tutoring.

What this paper adds

- ITSs produce a larger effect size on reading comprehension when compared to traditional instruction.
- ITSs for reading produce a small effect size when compared to human tutoring.
- The effect size of ITSs on reading comprehension is inflated when studies use researcher-designed tests.
- Our study provides some evidence that ITSs with strong intensity–duration may result in a large effect on reading comprehension.

Implications for practice and/or policy

- These findings are important to practitioners and policy makers interested in improving reading comprehension using high quality, consistent and accessible ITSs.
- When implementing ITSs in an academic environment, it should be used with more intensity and for longer duration.

Introduction

Reading comprehension presents a substantial challenge for students in K-12 classrooms (eg, National Assessment of Educational Progress, 2015). Evidence from classroom observations show that over 50% of teachers observed do not deliver evidence-based reading instruction to the diverse groups of students in their classrooms (eg, Binks-Cantrell, Washburn, Joshi, & Hougén, 2012; Bohn, Roehrig, & Pressley, 2004; Beerwinkle, Wijekumar, Walpole, & Aguis, 2018). Some studies provide evidence that human reading tutors improve children's reading comprehension (Bloom, 1984; Chi, Siler, Jeong, Yamauchi, & Hausmann, 2001; Cohen, Kulik & Kulik, 1982; Meyer *et al.*, 2002). Unfortunately, human tutoring can be expensive, inconsistent and difficult to scale up for large numbers of learners. However, intelligent tutoring systems (ITSs), a computer-based instructional system that provides immediate and customizable feedback to learners, is an alternative that delivers consistent and scalable instruction (Graesser, Conley, & Olney, 2011; Phobun & Vicheanpanya, 2010; Snow, Burns, & Griffin, 1998). A review conducted by VanLehn (2011) demonstrated that ITSs ($d = 0.76$), were as effective as human tutors ($d = 0.79$) at improving student academic outcomes across subject areas.

Previous reviews have focused on domains including science and mathematics (Fletcher-Flinn & Gravatt, 1995), reading comprehension intervention utilizing different formats (eg, Cheung & Slavin, 2012), or on specific instructional approaches to reading comprehension (eg, Hebert, Bohaty, Nelson, & Brown, 2016). However, there have been no systematic reviews focused on the impact of ITSs on reading comprehension specifically. To fill this gap, this meta-analysis investigated the effects of ITSs on reading comprehension for K-12 students.

Literature review*Intelligent tutoring systems (ITSs)*

ITSs are computerized learning environments that are designed to promote learning using complex computational models of student cognitive and motivational states. Using artificial

intelligence (AI) techniques, ITSs facilitate learning based on cognitive psychology principles and student models of learning (Anderson, Corbett, Koedinger, & Pelletier, 1995; Shute & Psotka, 1996; Steenbergen-Hu & Cooper, 2013). Being interactive is an important characteristic of ITSs. Shute and Zapata-Rivera (2008) adopted a four-process cycle of learner interaction: capture, analyze, select and present. Within this model, ITSs analyze an individual learner's needs by assessing or capturing his/her performance, select corresponding tasks and present appropriate information to the learner. ITSs is an interactive system. These systems use interactions between the learners and the system (Anderson, Corbett, Koedinger, & Pelletier, 1995; Atkinson, 1968) to make diagnoses about the student's knowledge of the content, strategy use, motivation and engagement. This information is then used to deliver timely and tailored feedback to learners, thus creating adaptive instructional input for individuals during tutoring sessions (Shute & Psotka, 1996).

With regard to reading instruction, ITSs usually include the following components: (1) modeling; (2) prediction; (3) feedback; (4) adaptive lessons and activities; and (5) scaffolding (McNamara, O'Reilly, Rowe, Boonthum, & Levinstein, 2007). Not all ITSs have all five components, but all of them track the cognitive state(s) of learners through a process called student modeling which represents the learners' needs (Graesser, Conley, & Olney, 2011; VanLehn, 2006). According to Graesser *et al.* (2011), although other computer programs adapt to individual learners this adaptation is not based on complex learning principles, nor do these programs aim to track the cognitive or emotional states of learners, two features of ITSs.

Previous reviews

There are several studies that have examined the effectiveness of educational technology on general reading outcomes. Blok, Oostdam, Otter, and Overmaat (2002) studied the effectiveness of computer-assisted instruction ($g = 0.19$) for beginning readers (K-3). More recently, Cheung and Slavin (2012) reviewed the impact of specific features of technology programs on reading outcomes for students in grades K-12. They found educational technology applications had a positive, though small, effect ($g = 0.15$). In 2013, Cheung and Slavin conducted an additional synthesis of the effects of educational technology applications on reading outcomes, this time focusing on struggling readers. The educational technology applications produced a positive but small effect on the reading skills of these readers ($g = 0.14$). This research on struggling and beginning readers is valuable but focusing on a broader group of students will benefit practitioners and researchers alike and assist them when making decisions about reading technology applications.

There have been several meta-analyses examining the effectiveness of ITSs for student learning outcomes. Steenbergen-Hu and Cooper (2013) conducted a meta-analysis investigating the role of ITSs on K-12 students' mathematical learning. They found that ITSs had a small positive effect (from $g = 0.01$ to $g = 0.09$) on students' mathematical learning. Steenbergen-Hu and Cooper (2014) later conducted a meta-analysis measuring the effectiveness of ITSs on college students' academic learning and found that ITSs had a moderate positive effect on academic learning (from $g = 0.32$ to $g = 0.37$). Ma, Adesope, Nesbit, and Liu (2014) conducted a meta-analysis comparing student learning outcomes for students using ITSs to students not using ITSs. They found that the ITS group performed better than the other three control conditions: the effect size was 0.42 when compared to large-group instruction, 0.57 when compared to non-ITSs computer-based instruction and 0.35 when compared to the textbook/workbook group. However, there was no significant difference between learning from ITSs and learning from individualized human tutoring ($g = -0.11$) or small-group instruction ($g = 0.05$). More recently, Kulik and Fletcher (2016)

conducted a meta-analysis on the effectiveness of ITSs as compared to traditional instruction. They found that the median effect of ITSs was an increase of 0.66 of a standard deviation on students' test scores. These meta-analyses add to the field's understanding of the facilitative role of ITSs in the classroom, but they did not consider the impact of ITSs on reading comprehension. While there have been a number of meta-analyses about the effectiveness of ITSs or a number of meta-analyses about the effectiveness of educational technology on general reading outcomes, none of them have considered the impact of ITSs on reading comprehension specifically. Due to the role of reading comprehension in all academic activities understanding how to best deliver effective instruction in this area is paramount. Given findings of recent research, we hypothesize that ITSs may be more successful at improving reading comprehension outcomes than traditional instruction and thus a review focusing primarily on ITSs and reading comprehension is justified and needed.

Research questions

This meta-analysis examines the effectiveness of ITSs on reading comprehension. It is designed to answer the following research questions:

1. Do ITSs improve reading comprehension for K-12 students more effectively than other instructional methods?
2. How do studies' substantive and methodological characteristics, such as program intensity and duration, and research design, impact the estimated effects?

Method

Study search

To find studies for analysis, we conducted computer searches of databases from four resources: (a) the Educational Resources Information Clearinghouse (ERIC), (b) Academic Search Complete, (c) Computer Source and (d) PsycINFO. The search included ITSs, as a keyword concept, specific ITSs names and reading comprehension. For each of these searches, thesaurus terms were searched along with keywords in the titles and abstracts to find all of the relevant citations. The search protocol in ERIC was limited to peer-reviewed articles published between 2000 and 2017. The following keywords were used: Intelligent Tutoring Systems, computer assisted, artificial, web-based, comprehension, reading, etc. The search was then translated for the other databases. We then conducted a manual examination of the references and bibliography lists of the relevant studies. No additional publications were identified during this step.

Criteria for inclusion

Included studies met the following eight criteria.

1. An empirical investigation of the effects of ITSs on reading comprehension instruction provided in a student's first language. ITSs developed for second language learners were excluded. However, if ITSs was developed to teach a language other than English, it was included in the study.
2. Published in a peer-reviewed journal during the time period from January 1, 2000 to March 31, 2017 and available in English. We delimited our search to this time period because the number and functionality of ITSs have changed since 2000. Secondary data analyses, literature reviews, dissertations and conference papers were excluded.

3. Samples included high achieving, low achieving and/or low socio-economic status (SES) students in grades K-12.
4. Examined the effectiveness of ITSs on reading comprehension as measured by standardized tests and/or researcher-designed tests.
5. Included an independent comparison group. Comparison conditions included traditional teacher-led classroom instruction (coded "TI"), human tutoring (coded "HT") or other types of educational programs (coded "EP"). However, publications that focused on comparing an ITSs with an updated version of that ITSs were excluded.
6. Employed randomized experimental or quasi-experimental designs.
7. Provided the necessary quantitative information for the calculation or estimation of effect size.

Study coding

Due to the small number of studies included in the meta-analyses, we limited our analysis to four predictor variables and examined their contribution to between-group heterogeneity. The variables analyzed were: outcome measures for reading comprehension, comparison group, the interaction between program intensity and duration and research design.

Previous studies indicate diversified effects for technology's intensity and duration on literacy (eg, Cheung & Slavin, 2013; Xu, Banerjee, Ramirez, Zhu & Wijekumar, 2018). So for this meta-analysis, we simultaneously explored the impact of duration and intensity of different ITS interventions so that we could determine how to best maximize its effect on reading. We followed Cheung and Slavin's (2013) criteria for intensity, with a cutoff of 75 minutes per week. In addition, we argue that duration of ITSs may play a vital role in the possible effects on reading comprehension outcomes. Therefore, we added duration as a measurement outcome, with a cutoff of three months' length. We used three months because that timeframe represents at least half of a typical school year of 180 days. We coded an ITS as strong when it was used more than three months and with an intensity of more than 75 minutes a week. Otherwise, it was categorized as weak (see Figure 1).

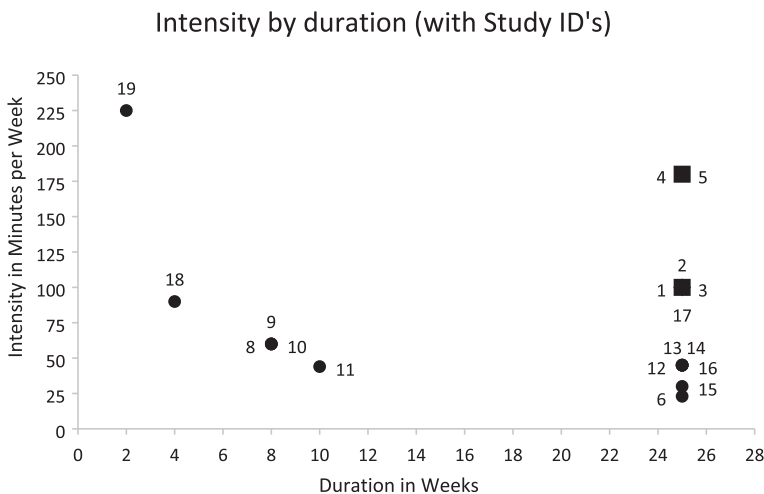


Figure 1: Intensity by duration (with study ID's)

We also focused on whether the effect of ITSs on reading comprehension varied based on reading comprehension outcome measures, comparison group and/or research design. This analysis produced meaningful guidance for researchers and practitioners and we hope will encourage additional systematic research on improving reading comprehension using ITSs in the future.

Two types of outcome measures were used in this analysis—standardized test measures and researcher-designed measures. A standardized test is any examination that is administered and scored in a predetermined, standard manner. The standardized tests included in this meta-analysis were: Woodcock Reading Mastery Tests (WRMT) (eg, Chalmers *et al.*, 2008a; Mostow, Nelson-Taylor, & Beck, 2013), Colorado Student Assessment Program (CSAP) (eg, Franzke, Kintsch, Caccamise, Johnson, & Dooley, 2005) and the Gray Silent Reading Test (GSRT) (eg, Wijekumar, Meyer, & Lei, 2012, 2013, 2017; Wijekumar *et al.* 2014). Researcher-designed measures were often proximal measures of learning developed by the researchers, which assessed knowledge of the constructs that were taught within the intervention. For example, in Wijekumar's study (2012), researchers developed a cloze signaling task, for which participants were required to generate signaling words when blanks appeared in comparative texts and a main idea quality task, for which participants were required to use a comparison structure text to write a two-sentence main idea. Both of the tasks were used to examine participants' proficiency with the reading skills instructed by the software.

We categorized the comparison groups into human tutoring group, traditional instruction group and other educational programs group. Human tutoring refers to tutoring by human beings instead of computer-assisted tutoring programs. For instance, in the study conducted by Chambers *et al.* (2011), two human tutors were assigned to help a group of struggling readers. The human tutors performed the same assessment, planning and student activities and focused on the same objectives as the computer-assisted tutoring program Alphonie's Alley. Traditional instruction refers to normal classroom instructions (business-as-usual) at school without extra intervention. Other education programs refer to programs other than ITSs. For instance, in the study conducted by Lenhard *et al.* (2013), they compared the conText (an ITSs) with a teacher-directed instructional program, Reading Detectives. In Mostow *et al.* (2013), they compared outcomes between Project Listen's Reading Tutor (an ITSs) and Sustained Silent Reading.

The major features of each study were coded by two independent coders. The coders met to check agreement of the coding. When there was a dispute in coding, the two coders reevaluated the studies and decided which code was the most appropriate. They reached 95% agreement across all features. For those cases when agreement was not reached, the second author met with the two coders to resolve the issue, resulting in 100% agreement.

Effect size calculation

We used procedures described by Lipsey and Wilson (2001) to calculate an unbiased effect size (Hedge's *g*) for ITSs on reading comprehension for each study based on reported means and standard deviations. In the absence of reported means or standard deviations, we employed other procedures prescribed by Lipsey and Wilson (2001) using *p*-values, test statistics (*t* or *F*) and/or confidence intervals to estimate an effect size. For example, for Chambers and colleagues (2011), we used the posttest mean and *p*-value to obtain the effect size. In another study (Jackson, Varner, Boonthum-Denecke, & McNamara, 2013), we used a graph of means and confidence intervals to estimate the effect size. One study (Wijekumar, Meyer, & Lei, 2012) was designed for class level analysis, so we converted the effect size from class level to student level for convenience

of comparison since the rest of the studies reported their statistics at the student level (Higgins & Green, 2011, specific equations and calculation information are provided in Appendix A).

Statistical analysis

As a parametric procedure, meta-analysis is subject to underlying assumptions; one assumption is that the effect sizes being integrated are statistically independent. Many studies report more than one effect size and because these multiple effect sizes within a single study are not statistically independent, previous meta-analyses had to choose one effect size or average the multiple effect size estimates. A new method has been developed as an approach for integrating multiple effect sizes within a single study rather than choosing one or using their average: robust variance estimation (RVE).

RVE is a method for integrating statistically dependent effect sizes in meta-analysis (Hedges, Tipton, & Johnson, 2010; Tanner-Smith & Tipton, 2014). RVE allows researchers conducting meta-analyses to attend to statistical dependences arising from correlated estimation errors and/or correlated effect size parameters (Hedges, Tipton, & Johnson, 2010). In the former, statistical dependency of estimated errors may arise in connection with multiple estimates from the same sample (such as multiple measures of the same outcome), or partially overlapping samples (such as a shared control group). In the latter, statistical dependency may arise from including studies from the same laboratory, same investigator or team or studies reported in the same publication. In this review, we focus on the first source of statistical dependency and studies which included multiple outcome measures for reading comprehension.

Results

In total, there were 19 independent studies based on 19 independent samples taken from 13 articles, from which 88 effect sizes were extracted. In the 19 studies analyzed for this meta-analysis 6 ITS programs were used: Alphie's Alley, conText, iStart, ITSS, Project Listen and Summary Street. Two of them are based on latent semantic analysis (LSA): conText is a computer-assisted program that helps students grasp the meaning of a text by supplying guided practice (Lenhard *et al.*, 2013) and Summary Street is a comprehension and writing tutor which provides an instantaneous evaluation of students' summaries (Landauer, 2002; Landauer & Dumais, 1997). Some ITSs are based on a reading strategy. For instance, Intelligent Tutoring of the Structure Strategy (ITSS) is a web-based tutoring system which facilitates understanding of text by explicitly teaching learners how to follow the propositional structure of text by applying knowledge about text structures (Meyer *et al.*, 2010; Meyer, Young, & Bartlett, 1989). There are also ITSs designed to help human tutors, rather than replacing them entirely (Danis *et al.*, 2005; Mitchell & Grogono, 1993). Alphie's Alley is one such software program. It provides assessment, planning tools and performance support for reading tutors (Chambers, Abrami, *et al.*, 2008, 2011).

As previously noted, the ITSs included in this meta-analysis do not contain all of the components of an intelligent tutoring system—modeling, prediction, feedback, adaptive lessons and activities and scaffolding. But all of them employ student modeling, which tracks the cognitive state(s) of learners and adapts to individual learners' needs (detailed information about the components of the included ITSs is in Table 1). For example, Alphie's Alley provides consistent feedback and scaffolding based on students' performance by tracking students' progress and accommodating students' diversity (Chambers, Abrami, *et al.*, 2008) and ITSS is designed to provide modeling, interactive practice, assessment and feedback to students so that they learn how to use text structure strategy (Meyer *et al.*, 2002).

Table 1: ITSs components

ITSs included in the meta-analysis	Modelling	Prediction	Feedback	Adaptive lessons and activities	Scaffolding
Alphie's Alley	y	y	y	y	y
ConText	y	y	y	/	/
iSTART	y	y	y	y	y
ITSS	y	y	y	y	y
Project LISTEN's Reading Tutor	y	/	y	y	/
Summary Street	y	/	y	/	/
TuinLEC	y	y	y	y	y

Note. y = component covered; / = component not covered.

Among the 19 studies, 12 compared the effect of ITSs with traditional instruction, 5 investigated the difference between ITSs and human tutoring and 2 researched the difference between ITSs and other educational programs. Across all of the studies, there were approximately 10 000 students from grades 1 through 10 (although we sought to include studies with students in K-12th grade we did not find studies for that range that met our inclusion criteria), with each study's sample size ranging from 25 students to more than 2000 students. Among them, four manuscripts were published in the 2000s and nine were published in the 2010s. Seventeen studies employed a randomized experimental design whereas the other two used a quasi-experimental study design. The intensity of the programs varied from 20 to 225 minutes per week. The duration of the programs ranged from two weeks to an entire school year. Table 2 summarizes the basic features and results of the qualifying studies.

Summary estimates using RVE

The overall summary estimate of the 88 effect sizes was 0.60 (95% CI = 0.36–0.85) with a *p*-value of 0.0001.

Summary estimate controlling for four predictors

According to Tanner-Smith and Tipton (2014), the number of moderators that can be included in sub-group analysis is limited by the number of studies included, not by the number of effect sizes. For this reason, we conducted four sub-group analyses: outcome measures for reading comprehension, comparison group, the interaction between program intensity and duration and research design.

Meta-regression results controlling separately for each of the four predictor variables are presented in Table 3. In each regression model, the “beta” is a measure of the overall effect size (unadjusted in Model 1) and the effect size adjustment for the predictor variables (Models 2–5). In all five models, effect size was statistically significant; and three of the predictor variables were statistically significant: outcome measures ($p = 0.0013$), comparison group ($p = 0.0046$) and the interaction between intensity and duration ($p = 0.0097$). The summary effect sizes for each of the statistically significant predictor variables are presented in Figure 2.

Subsequent analyses

In order to explore multivariate relationships (eg, outcome measures, comparison group, intensity–duration interaction), and to avoid multicollinearity in the meta-regression, bivariate correlation analyses were carried out using the spearman correlation because all of the independent variables were dichotomous and categorical (See Table 4).

Table 2: Basic features and effect size information for all studies

Publication/ Study ID	Study	Comparison group	Intensity * duration	Sample-n	Grade level	Measures outcome	Research design	# of ESs	ES
<i>ITSs name: Alphie's Alley</i>									
1.1(1)	Chambers, Slavin, <i>et</i> <i>al.</i> (2008)	HT	1 school year/100 minutes a week	412	1	ST	TE	5	0.04
1.2(2)	Chambers, Slavin, <i>et</i> <i>al.</i> (2008)	HT	1 school year/100 minutes a week	203	1	ST	TE	5	0.14
2(3)	Chambers, Abrami, <i>et al.</i> (2008)	HT	1 school year/100 minutes a week	60	1	ST	QE	5	0.42
3.1(4)	Chambers <i>et</i> <i>al.</i> (2011)	HT	1 school year/180 minutes a week	366	1	ST	TE	3	0.24
3.2(5)	Chambers <i>et</i> <i>al.</i> (2011)	HT	1 school year/180 minutes a week	280	2	ST	TE	3	0.15
<i>ITSs name: conText</i>									
4(6)	Lenhard, <i>et</i> <i>al.</i> (2013)	EP	1 school year/45 minutes two week	139	6	ST	TE	4	0.04
<i>ITSs name: iSTART</i>									
5(7)	McNamara, O'Reilly, Best, and Ozuru (2006)	TI	whole training lasted an average of 1 hour 44.4 minutes	39	8, 9	RT	QE	3	0.29
6.1(8)	Jackson <i>et al.</i> (2013)	TI	eight 1-hour sessions	40	10	RT	TE	1	1.43
6.2(9)	Jackson <i>et al.</i> (2013)	TI	eight 1-hour sessions	38	10	RT	TE	1	1.38
6.3(10)	Jackson <i>et al.</i> (2013)	TI	eight 1-hour sessions	40	10	RT	TE	1	1.06
6.4(11)	Jackson <i>et al.</i> (2013)	TI	eight 1-hour sessions	44	10	RT	TE	1	1.46

Table 2: Continued

Publication/ Study ID	Study	Comparison group	Intensity * duration	Sample-n	Grade level	Measures outcome	Research design	# of ESs	ES
<i>ITSs name: ITSS</i>									
7(12)	Wijekumar <i>et al.</i> (2012)	TI	6 months/30–45 minutes a week	129 classes /2643 students	4	ST+RT	TE	7	0.62
8.1(13)	Wijekumar <i>et al.</i> (2013)	TI	6–7 months/30–45 minutes a week	24 classes/437 students	4	ST+RT	TE	5	1.21
8.2(14)	Wijekumar <i>et al.</i> (2013)	TI	6–7 months/30–45 minutes a week	24 class/413 students	5	ST+RT	TE	5	1.55
9(15)	Wijekumar <i>et al.</i> (2014)	TI	1 school year/20–30 minutes a week	130 classes/2485 students	5	ST+RT	TE	7	0.31
10(16)	Wijekumar <i>et al.</i> (2017)	TI	1 school year/30–45 minutes a week	108 classes/2024 students	7	ST+RT	TE	13	0.39
<i>ITSs name: Project LISTEN's Reading Tutor</i>									
11(17)	Mostow <i>et al.</i> (2013)	EP	7 months/100 minutes a week	178	1–4 grade	ST	TE	12	0.38
<i>ITSs name: Summary Street</i>									
12(18)	Franzke <i>et al.</i> (2005)	TI	4 weeks/90 minutes a week	111	8	ST+RT	TE	6	0.19
<i>ITSs name: TuinLEC</i>									
13(19)	Vidal-Abarca <i>et al.</i> (2014)	TI	2 weeks/225 minutes a week	25	6	ST	TE	1	0.69

Note. TI = traditional instruction; HT = human tutoring; EP = other educational programs; ST = standardized test; RT = researcher-designed test; TE = true experimental; QE = quasi-experimental.

Table 3: Meta-regression analyses

Model	Regression Component	Beta	SE	df	p-value	95% CI	τ^2
1	Constant**	0.604	0.118	17.52	0.0001	0.355 to 0.852	0.246
2	Outcome Measures*	0.736	0.191	16.63	0.0013	0.333 to 1.139	0.223
	Constant**	0.258	0.054	9.43	0.0008	0.138 to 0.379	
3	Comparison Group*	-0.651	0.175	9.27	0.0046	-1.045 to -0.256	0.245
	Constant**	0.499	0.080	9.79	0.0001	0.321 to 0.678	
4	Research Design***	-0.267	0.146	1.211	0.2831	-1.504 to 0.970	0.248
	Constant**	0.628	0.129	15.580	0.0002	0.354 to 0.903	
5	Intensity–Duration*	0.523	0.166	10.26	0.0097	0.158 to 0.896	0.248
	Constant*	0.256	0.059	4.98	0.0073	0.105 to 0.407	

Notes. Statistical Significance: * $p < 0.01$, ** $p < 0.001$, *** p -value untrustworthy (degrees of freedom < 4). Control variable codes: Type of Measure: 0 = standardized; 1 = researcher-designed; Comparison Group: -0.5 = traditional instruction; 0 = other educational program; 0.5 = human tutoring; Research Design: 0 = true experiment; 1 = quasi-experiment; Intensity–Duration: 0 = strong; 1 = weak.

Highly correlated bivariate relationships exist among these data for comparison group and the intensity–duration interaction ($\rho = -0.9209$), comparison group and outcome measures ($\rho = -0.7329$) and outcome measures and intensity–duration interaction ($\rho = 0.6911$). Correlation matrices can be deceiving when using dichotomous/categorical data, so we examined distribution of effect sizes across substrata instead (See Figure 3).

We also looked at the bivariate effect size by outcome measures, comparison group and intensity–duration interaction (See Table 5). When ITSs were compared to traditional instruction of weak intensity–duration, researcher-designed measures resulted in inflated effect sizes (1.01), approximately four times the standardized measure effect size (0.26). When compared to other educational programs, strong intensity–duration (0.37) resulted in a much larger effect than weak intensity–duration (0.04) (Note: these two summary effect sizes were computed using traditional meta-analytic methods, not RVE, because each stratum represented a single study with

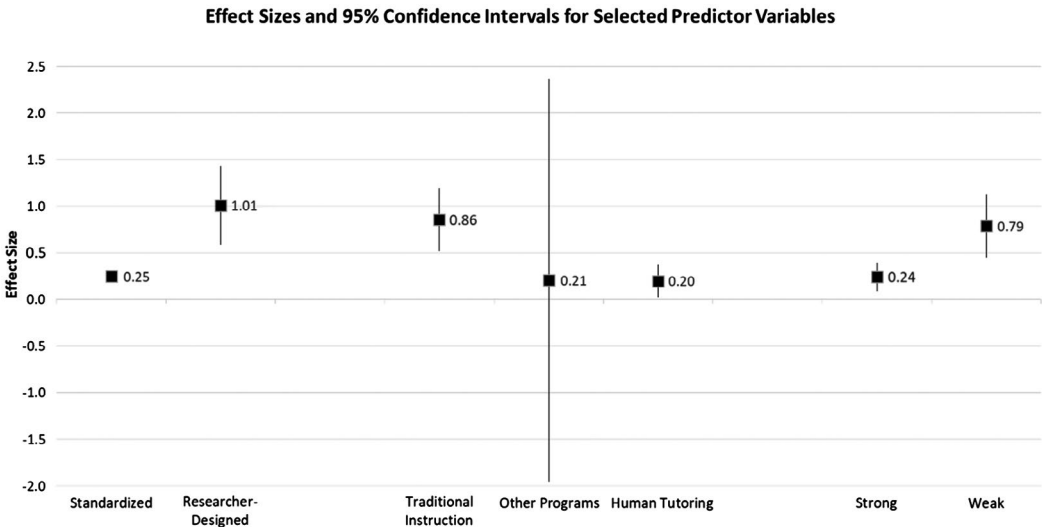


Figure 2: Effect sizes and 95% confidence intervals for selected predictor variables

multiple effect sizes each. Then the effect sizes across both cells were combined using RVE, resulting in a combination of strong and weak intensity–duration for standardized measures; the result was statistically not significant when combined. Thus, the relationship between strong and weak intensity–duration among standardized measures only should be interpreted with caution). ITSs compared to other programs of strong intensity–duration (0.37) produced a larger effect than when compared to human tutoring of strong intensity–duration (0.20).

Publication bias analyses

A funnel plot is a visual aid for detecting bias (typically publication or other reporting bias) or systematic heterogeneity; it is constructed as a scatter plot of effect sizes against study precision (standard error). In the absence of bias or systematic heterogeneity, the data will be distributed throughout a symmetric inverted funnel. The funnel plot in Figure 4, with its voids, suggests a potential publication/reporting bias or systematic heterogeneity, which should be considered when interpreting the results of this meta-analysis.

Discussion

Overall effectiveness of ITSs for reading comprehension

This meta-analysis provides information about the effectiveness of ITSs for improving reading comprehension for first through tenth-grade students. To the best of our knowledge, this is the first study to examine the effect of ITSs on reading comprehension specifically. Because the studies included in our analysis used more than one measure of reading comprehension, we employed an RVE approach for analysis. This approach enabled us to examine and compare effect sizes when using standardized and researcher designed measures of reading comprehension. We identified 19 independent studies in 13 peer-reviewed publications that met our criteria. An effect size of 0.2 is small, above 0.5 is medium and above 0.8 is large (Cohen, 1988). Our results indicate that ITSs had a medium effect ($g = 0.60$) on students' reading comprehension. However, if we focus on the effect size when only considering standardized outcome measures, the effect size is small (0.25). This may be due to the distal nature of the standardized measures to the intervention as opposed to the proximal nature of the researcher-designed measures.

The findings of this meta-analysis indicated that ITSs produced larger effects than traditional instruction and other educational applications. When compared to traditional instruction, ITSs produced a large effect size (0.86). Analyzing the researcher-designed measures and standardized measures separately, the effect size was 1.01 and 0.27 respectively.

This meta-analysis confirmed findings from previous studies comparing ITSs to human tutoring. We found that ITSs produced a small but positive effect when compared to human tutoring (0.20,

Table 4: Correlation matrix

Variables	ES	Outcome measures	Comparison group	Intensity–Duration	Research design
ES	1.00				
Outcome measures	0.50	1.00			
Comparison group	−0.37	−0.73	1.00		
Intensity–duration	0.31	0.69	−0.92	1.00	
Research design	0.13	−0.04	0.20	−0.16	1.00

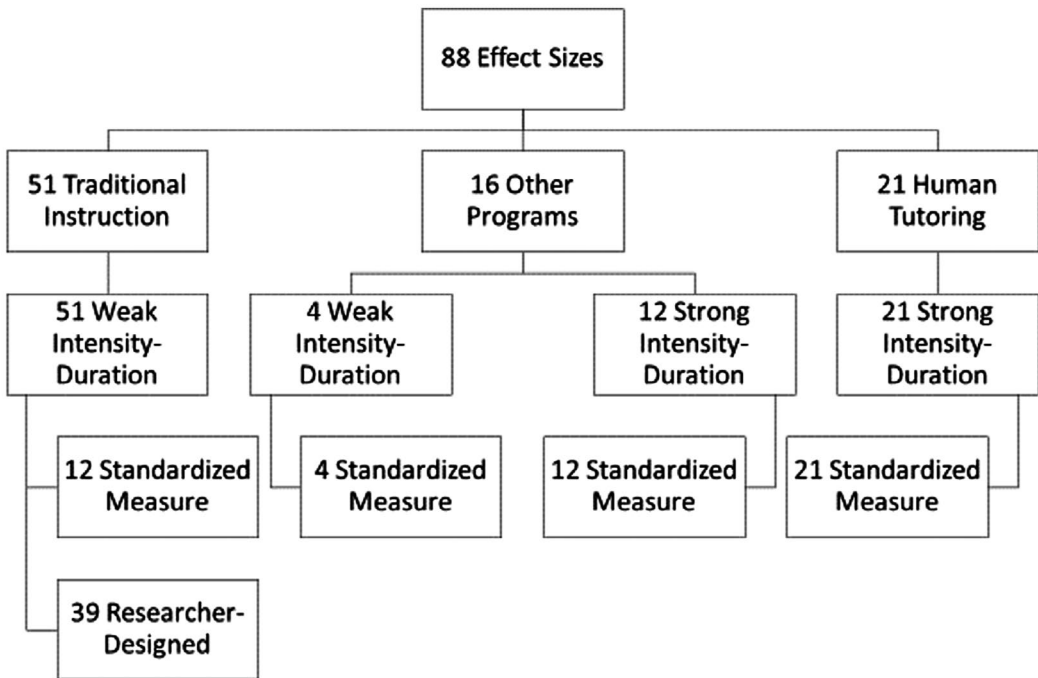


Figure 3: Distribution of effect sizes across substrata

using standardized measures), which is marginally greater than Ma *et al.* (2014)'s finding (0.11). In spite of the small effect size (0.20) when compared to human tutoring, our findings support previous research which has found that ITSs can be as effective as human tutoring (VanLehn, 2011), and therefore could be an effective tool to improve children's reading comprehension (Snow, Burns, & Griffin, 1998).

Findings from testing for predictors

Intensity–duration of ITSs

We hypothesized that ITSs with strong intensity and long duration would produce a larger effect size than ITSs with weaker intensity and shorter duration. However, surprisingly, the results contradicted this. Stronger ITSs yielded an effect size of 0.26 and weaker ITSs an effect size of 0.78 when using mixed measures ($p = 0.0097$). When only standardized measures were analyzed there was no significant difference between the strong and weak groups (0.23, and 0.26 respectively, $p = 0.60$). These results differ from those reported by Cheung and Slavin (2013), who found that compared to low-intensity educational technology applications, high-intensity programs had a bigger impact on struggling readers. Steenbergen-Hu and Cooper (2013) found that the effect size of an ITS intervention was larger if the ITS lasted for less than a school year as compared to those that lasted an entire school year. However, as previously noted when we combined intensity and duration, we found a contradictory result when using standardized and researcher designed measures and no significant difference when using standardized measures only. This result may be due to the fact that a majority of the weak programs were using researcher-designed tests ($n = 39$), which inflated the effect sizes of the weak programs.

Table 5: Bivariate effect size (95% confidence intervals; p-values, n of effect sizes) by outcome measures by comparison group and intensity-duration

Predictor variables and sub-groups	Outcome measures			
	Standardized		Researcher-designed	
	Strong intensity-duration	Weak intensity-duration	Weak intensity-duration	
Comparison group				
Traditional instruction	n/a	0.26 0.17 to 0.36 p = 0.0029, n = 12	1.01 0.58 to 1.43 p = 0.0005, n = 39 n/a	
Other program	0.37* 0.07 to 0.68 p = 0.0217, n = 12	0.04** -0.13 to 0.20 p = 0.6810, n = 4		
Human tutoring	0.20 0.02 to 0.38 p = 0.0364, n = 21	0.21*** -1.95 to 2.37 p = 0.4386, n = 16		

*12 effect sizes from a single publication/study;
**4 effect sizes from a single publication/study estimated using metan command (not RVE);
***All 16 effect sizes, strong and weak, combined using RVE.

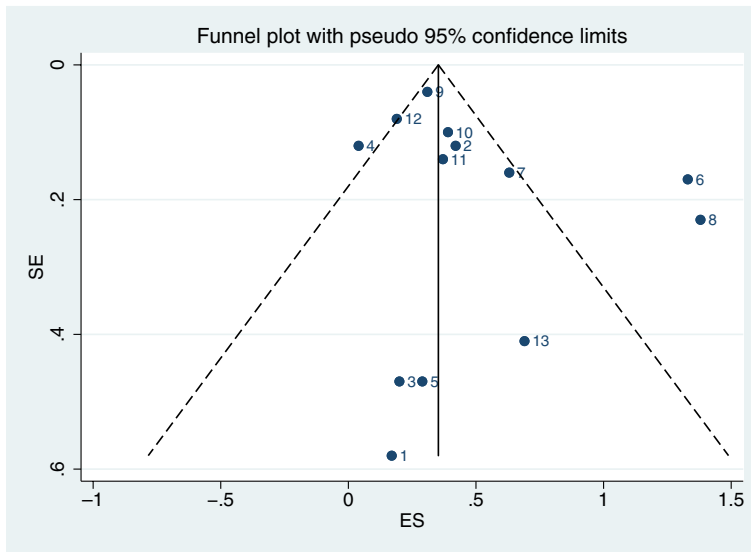


Figure 4: Funnel plot
[Colour figure can be viewed at wileyonlinelibrary.com]

Although our hypothesis regarding intensity and duration was not supported by the overall effect sizes, there were some observable and interesting findings. First, we found that when ITSs were compared to other educational programs—*Detectives* and *Sustained Silent Reading*, strong intensity–duration (0.37) resulted in a much larger and statistically significant effect than weak intensity–duration (0.04). *Detectives* is a teacher-directed strategy instruction program. *Sustained Silent Reading* is a reading program, during which students read silently without interruption. The effect sizes were measured by standardized tests across program-type so it was a like comparison. It should be noted that the confidence interval is wider for the other educational groups as compared to the comparison groups, which indicates greater substantive heterogeneity between the programs. Furthermore, these findings should be interpreted with caution because there were only two studies which compared ITSs to the other educational programs and these two programs differed from one another.

Researcher-designed measures

There were a greater number of researcher-designed measures used (39 out of 51 effect sizes). RVE suggests that there will be inflated effect sizes when using researcher designed measures which we observed in our data. The stratified analysis results for effect size for researcher-designed and standardized measures were 1.01 and 0.25 respectively. Thus, the effect size found using researcher-created assessments was approximately four times greater than that for standardized measures. This finding is consistent with Ma *et al.* (2014)'s findings that ITSs produced larger effect sizes on researcher-developed learning measures than standardized learning measures when a fixed-effect model was used (0.40 and 0.27 respectively).

The reason researcher-designed measures produced higher effect sizes than standardized tests may be due to alignment of the measure with the knowledge and skills emphasized in the intervention. Although the researcher-designed measures serve the purpose of testing the effectiveness

of the intervention, future research may need to employ both measures to better understand distal effects.

Implications for practice

This meta-analysis confirmed previous studies of the effect sizes of ITSs compared with human tutoring (0.20) on student learning outcomes. Our findings confirmed that ITSs are a promising form of intervention because they can yield similar effects to human tutoring on reading comprehension (0.20), but it should be noted that to achieve this it may be necessary to implement an ITS of stronger intensity and longer duration than its human one-on-one intervention counterpart. However, this study also showed that ITSs produce an even larger effect size when compared with traditional instruction (0.86 and 0.27 using mixed measures and standardized measures only respectively) and other educational programs (0.21 using only standardized measures). These findings offer practical implications for practitioners and policy makers. Based on the prevalence of computers and improved Internet connectivity in schools, ITSs present a viable option for providing consistent and accessible instruction for reading comprehension. ITSs may provide a good opportunity for under-served and geographically dispersed populations who may otherwise not have easy access to well-trained teachers. We recommend that more researchers quantify the costs associated with an ITSs (eg, purchase price, set-up, implementation, maintenance, etc.) so that we can conduct a cost-benefit analysis of ITSs vs. other educational programs including traditional teacher-led instruction.

The current study also provides valuable insight into factors that affect the use of ITSs by introducing a consideration of intensity and duration. The results of this meta-analysis show that ITSs can potentially produce stronger results when ITSs are compared to human tutoring (0.20) if ITSs are used for more than 75 minutes a week (intensity) and longer than 3 months (duration). Thus, when teachers and school policy makers make decisions, they need to take intensity and duration of ITSs into consideration.

Limitations and future research

As with any meta-analysis, the studies we chose to include and exclude, coding procedures and predictor analysis restrict our findings. Because this study only includes peer-reviewed articles, there may be other studies relevant to this topic not identified. As the funnel plot shows, there might be publication bias since we excluded studies not published which may include studies with small or negative effect sizes. Though peer-reviewed publications guarantee the quality of the studies included, the current synthesis is limited by the inclusion and exclusion parameters we adopted.

We included a limited number of studies which confined our ability to draw conclusions. Only 13 publications met our standards, from those 19 independent studies were included in the meta-analysis. Limited by the number of studies included in the current meta-analysis, we only analyzed four predictors—outcome measures for reading comprehension, comparison group, program intensity–duration and research design. Additional moderators, such as sub populations (eg, low-performing students, English learners), combined effects of reading and writing outcomes and specific features of ITSs, should be considered in future studies. Also, ITSs are instructional interventions that should also be compared with other programs that do not use technology, eg, reading recovery (Shanahan & Barr, 1995). Though two of our included studies compared the effect of ITSs with other educational programs, this comparison could be extended in the future research. When there are more studies available on the effectiveness of ITSs on reading comprehension, future researchers may conduct a more comprehensive meta-analysis.

Finally, while conducting this meta-analysis, we noticed that there were some empirical studies comparing traditional ITSs with updated ITSs, such as game-based, individualized and ITSs implemented with high fidelity. Those studies were excluded in this meta-analysis since the current study only sought to compare ITSs with the instructional methods without ITSs. However, these studies provide evidence on how ITSs have developed into a mature tool for instruction.

Statements on open data, ethics and conflict of interest

The data is available from open databases—ERIC, Academic Search Complete, Computer Source, and PsycINFO.

This study was carried out under the ethical guidelines.

We declare have no conflicts of interest with respect to the research, authorship and/or publication of this work.

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APPENDIX A

Effect Sizes Conversion at the Cluster Level

Wijekumar, K. K., Meyer, B. J., & Lei, P. (2012) reported the data on class level. When clusters are the unit of experiment, we cannot compare the reported statistics with the data on students' level. To adjust for this, we followed Higgins and Green (2011) to obtain the sample size for treatment group, control group, and then calculate the effect size.

$$ICC = \frac{\text{variance of class}}{\text{total variance}}$$

DEFF = $1 + (M - 1) \times ICC$, where M is equal to the number of individual participants divide by the number of classes.

$$ESS = \frac{N}{DEFF}, \text{ where } N \text{ is equal to the number of individual participants in treatment or control group.}$$